

Cenhao (Claude) Li

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Education

Washington University in St. Louis	M.S. in Computer Science	GPA: 3.81/4.0	Sep 2021 – May 2023
South China University of Technology	B.S. in Computer Science	GPA: 3.64/4.0	Sep 2017 – July 2021

Professional Experience

Aurora <i>C++, Python, Bazel, protobuf, CUDA, gRPC, ZMQ</i> <i>Software Engineer</i> ◦ Autonomy Frameworks - Developed core framework runs on the autonomous vehicle(AV) and cloud infrastructure in C++ and Python, enhancing platform scalability and system robustness. - Engineered a low-latency Pub/Sub middle-ware system for seamless synchronization between modular autonomy components, and leveraged gRPC to manage distributed compute nodes. ◦ Motion Planner Architecture - Developed robust data abstraction layers in C++ that integrated online map and sensor streams, utilizing protobuf to serialize data for downstream planning algorithms, optimizing end-to-end planner module latency. - Improved Planner Ranker's recall and throughput by transitioning to graph-based lane representation and strategy-free boundary, preserving broader map context while eliminating redundant computation and storage.	Sep 2024 – present Mountain View, CA
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Hughes Network System <i>C++, CMake, CTest, ExpressCpp, ZMQ</i> <i>MTS2-Software Engineer</i> ◦ Architected and implemented robust FPGA interfaces in C++ to interact with 10+ PCIe devices on the satellite user terminal, a real-time embedded system, conducted unit-test with CTest , increasing code coverage from 60% to 90%. ◦ Elevated hardware communication and monitoring concurrency by optimizing synchronization and memory mechanisms to achieve a 10ms responsiveness target for mobile features, a reduction from 100ms in stationary counterparts. ◦ Developed and tested REST APIs using ExpressCpp , enabling efficient access to over 130 device statuses and streamlining task management through ZeroMQ , fostering seamless system integration with cross-functional teams.	July 2023 – Aug 2024 Gaithersburg, MD
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Internship

WeRide.ai <i>C++, Python, Bazel, Grafana</i> <i>Software Engineer Intern</i> ◦ Integrated safety module of the perception DAG data pipeline with C++ and Python to detect and handle over 40 fault scenarios, reduce 17% of false alarms and record informatic fault history efficiently with limited hardware resources. ◦ Enabled 3 fault injections to trigger sensor input, intra-module, and system-level failures in the simulation environment, and leveraged Grafana to visualize system metrics to validate fault injection results and identify potential issues. ◦ Developed diagnostic program monitoring GPU with NVML and recording delay accounting to triage 3-level failures. ◦ Extended Failure Mode and Effects Analysis (FMEA) scenarios from 14 to over 200 for the autonomous driving system.	May 2022 – Aug 2022 San Jose, CA
Intel Asia-Pacific R&D <i>Python, Bash, Docker, TensorFlow, PyTorch</i> <i>Software Engineer Intern</i> ◦ Built out-of-box scripts in Python to evaluate the performance of 6 TensorFlow and PyTorch variants of Intel and Nvidia. ◦ Employed Docker for containerized environments, enabling reproducible testing across 30+ ML workloads on each framework. ◦ Enabled the throughput of Keras workloads that under tf1.x in FP32 improve 12%-25% under intel-tf2.4 in BF16.	Mar 2021 – May 2021 Shanghai, China
Tencent <i>C++, Bash, MySQL, gRPC</i> <i>Software Engineer Intern</i> ◦ Developed scalable gRPC services in C++ to efficiently clean overdue records from large, continuously growing tables. ◦ Parallelized the cleaning process and optimized MySQL , reducing average execution time by 70% during large-scale deletions.	June 2020 - July 2020 Guangzhou, China

Skills

Programming Languages: C++, Python, CUDA, Java, Go, Bash, SQL
ML Frameworks & Tools: PyTorch, TensorFlow, Triton, NVML.
Other Tools: gRPC, ZMQ, (No)SQL, Docker, Git, Linux, Bazel, CMake, Protobuf, Grafana, Kafka.